

Electrode Digitization in EEG7 (A138)

Check beforehand

- Is workstation in EEG7 (around the corner to the right) switched on? → ... / ...

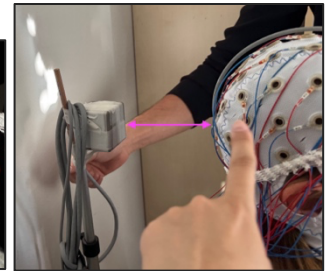
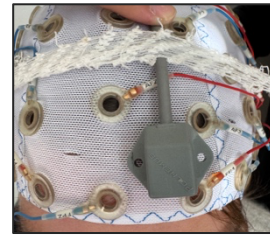
Requirements

- Participant has EEG cap positioned (nasion/inion checked + earlobes checked)
- Participant carries JackBox (like a present)

→ Guide to EEG7

Attach:

- **Sensor** onto head (in between: FPz, FP1, Afz, AF3)
→ Fixate with **white net!**
- Check: **Receiver Box** should be on the level of the scalp!



From now on, Person should not move anymore!

Procedure

- (1) Open Matlab R2007b (Click!)
- (2) Write into Command Window: **brainstorm**
[Brainstorm 3.2 opens)
- (3) **Brainstorm 3.2 Interface**
 - Click: File (top left) → Load protocol ...
→ **rsAirflow_Study1_Lioba**
 - Click: File (top left) → **Digitize**
 - GUI Opens → Enter Participant ID:
→ **sub-0XX** (with 0XX == Participant ID [001-050])
 - Click: **EEG montage** → **NeurOne_Easycap63_NTDetectMechanic**

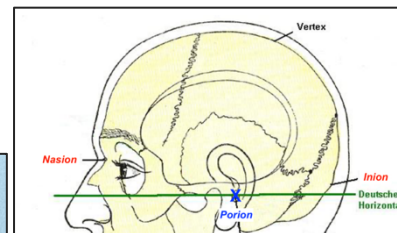
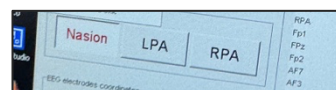
- (4) Use **Pen** + Locate onto Position + Click **Button on Pen** → Location registered!

- (5) Those three positions need to be located precisely 2 TIMES:

- **Nasion (NA)**
- **LPA** (left earlobe / porion)
- **RPA** (right earlobe / porion)

Anything goes wrong

→ File → **Start Over**



- (6) Once that has worked...

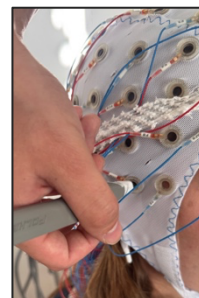
- **Person 1** reads out locations on screen
- **Person 2** uses pen to click within the electrodes to register the positions

EOG: Click on position **1 cm below the right eye!**

Don't click into GND (ground)!

Anything goes wrong →

Person 1 clicks on 'Delete last point'



- (7) Done with the electrode positions...

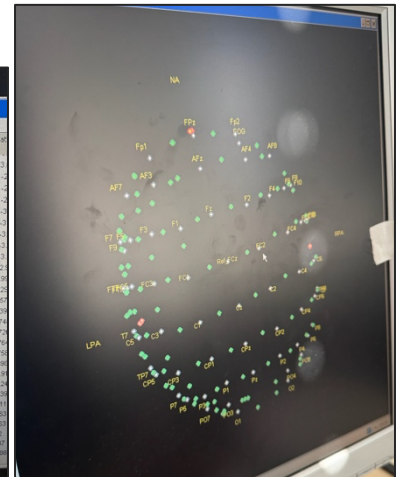
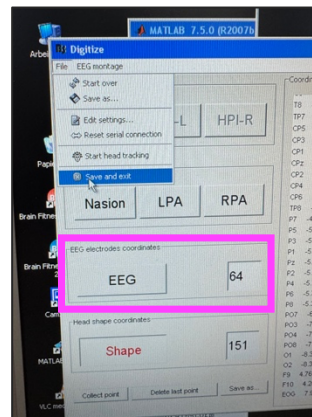
(8) Register Further Positions / Head Shape coordinates

- Each one lights up **Green**
- Move (1) along rows and (2) along columns

Goal: ~ N=150

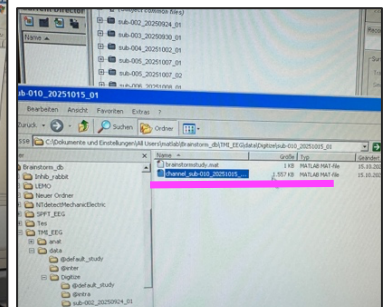
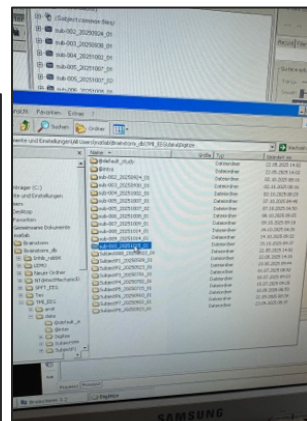
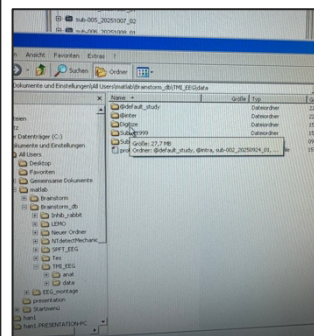
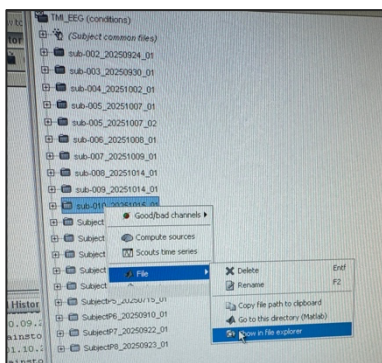
Done?

→ File → Save and exit



Check File Size

Click on **sub-0XX_Date** → File → Show in file explorer
 → Click on Folder **Digitize** → Click on File **sub-0XX_Date**:
File 'channel_...' should have about 1600 KB !!!



Data Transfer to EEG8 (→ CBS server)



Use the **USB stick 'Digitization USB'** for data transfer
 (located in right corner,
 in front of windows in EEG8)

Always return back to that location after usage!

Trouble shooting (!)

... when the digi pen doesn't work (doesn't make a beeping sound and/or no click is registered by the software):

[1] restart the computer

[2] restart the black box on the floor. "0" means off, and "I" means on.